Pratik Musale

Karlstad, Sweden | pratik.musale@kau.se | +46 72-186 85 42 | Personal Website | LinkedIn

Research Interests

Privacy and security in smart spaces and IoT ecosystems, with a focus on how users perceive, configure, and negotiate privacy in multi-user environments. My work combines large-scale surveys, qualitative methods, and system design to develop contextually-grounded privacy frameworks for smart homes and workplaces.

Education

University of Pittsburgh *PhD in Computer Science* August 2026
- Advisor: Dr. Adam Lee
- Thesis: Privacy in Smart Spaces: Exploring the Interplay Between User Preferences, Awareness Requirements, and Secure Technologies.

Stony Brook University *MS in Computer Science* June 2018
- Advisor: Dr. Bong Jun Choi
- Thesis: Lightweight Authentication for IoT Ecosystem

Savitribai Phule Pune University, India *BE in Electronics and Telecommunication* May 2015
- Advisor: Dr. Pranoti Mane
- Project: Accident Prevention System in Cars

Awards

Ph.D Fellowship, University of Pittsburgh Aug 2019-May 2020

Publications

[1] **Musale, Pratik** and Adam Lee. "Trust TEE?: Exploring the Impact of Trusted Execution Environments on Smart Home Privacy Norms". In: *Proceedings on Privacy Enhancing Technologies* (2023).

[2] Duin Baek, **Musale, Pratik**, and Jihoon Ryoo. "Walk to Show Your Identity: Gait-based Seamless User Authentication Framework Using Deep Neural Network". In: *The 5th ACM Workshop on Wearable Systems and Applications*. WearSys '19. Seoul, Republic of Korea: Association for Computing Machinery, 2019, pp. 53–58. ISBN: 9781450367752. DOI: 10.1145/3325424.3329666. URL: https://doi.org/10.1145/3325424.3329666.

[3] **Musale, Pratik**, Duin Baek, Nuwan Werellagama, Simon S. Woo, and Bong Jun Choi. "You Walk, We Authenticate: Lightweight Seamless Authentication Based on Gait in Wearable IoT Systems". In: *IEEE Access* 7 (2019), pp. 37883–37895. DOI: 10.1109/ACCESS.2019.2906663.

[4] **Musale, Pratik**, Duin Baek, and Bong Jun Choi. "Lightweight gait based authentication technique for IoT using subconscious level activities". In: *2018 IEEE 4th World Forum on Internet of Things (WF-IoT)*. 2018, pp. 564–567. DOI: 10.1109/WF-IoT.2018.8355210.

Manuscript Under Review or Plan of Submission

[1] **Musale, Pratik**, Gautam Aakash, and Lee Adam. "One Interface to Rule Them All?: Exploring Privacy Awareness Across Smart Workplace Spaces." Plan to Submit to CHI 2027.

[2] **Musale, Pratik**, Gautam Aakash, and Lee Adam, "From Administrative to Social: IoT Configurations in Workplace Meeting Rooms." Plan to Submit to CSCW 2027.

[3] **Musale, Pratik**, Gautam Aakash, and Lee Adam. "Bridging the Privacy Expectation Gap: Community-Aligned Defaults for Workplace IoT Configurations." Plan to Submit to PETS 2027.

Research

User-Aligned IoT Configurations in Workplace Environments 2024-Present

- Identified the *Community Coherence* phenomenon: users systematically prefer mid-ranked and least-negotiated configurations over privacy-centric or majority-voting defaults, reflecting social moderation over personal assertion.
- Showed that workplace configuration conflicts are anticipable rather than random, structured by three factors: privacy risk, utility benefit, and space alignment.
- Computed five algorithmic configuration defaults from group preferences across four IoT devices and three workplace scenarios (N=435).
- Tools: Qualtrics, IBM SPSS, Python

Group IoT Configuration Dynamics in Workplace Meeting Rooms 2024-Present

- Discovered that group IoT configuration produces conditional, activity-driven, and asymmetric policies that existing access control mechanisms cannot represent — a fundamental limitation with design implications.
- Identified two previously undescribed security risks: *policy collapse* and *configuration-induced insider exfiltration*; proposed Group CI-Profiles as a mechanism to preserve group-negotiated agreements.
- Found individual users rely on identification risk and limited data retention as primary configuration heuristics, distinct from group behavior.
- Conducted five focus groups (N=20) using CI-structured flashcard scenarios across video conferencing and in-person activities. Tools: NVivo

Privacy Awareness Interfaces for Smart Workplaces 2024-Present

- Found that data type and sensing purpose are consistently prioritized over other contextual integrity parameters across four workplace spaces and four IoT devices (N=399).
- Established space-specific interface preferences: users prefer dashboards for collaborative spaces and AR for individual spaces, driven by social norms and usage patterns.
- Designed and evaluated four interface prototypes (notice, push notification, dashboard, AR app) presenting CI information in a two-layer hierarchy; validated via 15 semi-structured interviews.
- Tools: Qualtrics, IBM SPSS, Python, NVivo, JavaScript, Android, Unity (AR)

Impacts of Trusted Execution Environments on Smart Home Privacy 2022-2023

- Demonstrated that secure hardware (TEE) reduces notification fatigue and simplifies users' mental models, but does not resolve bystander privacy concerns — showing that technical security alone cannot address multi-user smart space complexity.
- Found TEE collapsed the audio/video distinction in user mental models, with differential effects across device manufacturer and subject (user vs. bystander) conditions.
- Between-subjects survey of 1,049 American adults across smart speaker and camera scenarios. Tools: Qualtrics, IBM SPSS, Python

Prior Research

Lightweight Gait-Based Authentication for IoT (MS Thesis, Stony Brook University) 2016-2018

- Developed a lightweight continuous authentication system using gait patterns from smartphone and smartwatch sensors, reducing feature overhead while maintaining accuracy across 49 participants.
- Demonstrated feasibility of passive, sensor-based user authentication for IoT without specialty hardware; resulted in two IEEE publications and one ACM workshop paper.
- Tools: Android, Python, NoSQL, TensorFlow, SciKit-Learn

Appointments

Department of Computer Science, University of Pittsburgh
Graduate Teaching/Research Assistant June 2020-Present

Department of Computer Science, Stonybrook University
Graduate Teaching/Research Assistant Aug 2016-May 2018

Work Experience

Navigate Consulting - [Security Researcher Intern] Jan 2019-June 2019

- To investigate the feasibility of single sign-on feasibility for the Industry IoT ecosystem.
- Designing prototypes for the deployment of IoT devices in the field for a metal fabrication plant.

Spectra Televentures - [OMCR Engineer] Sep 2015-Jul 2016

- Maintenance of Radio Towers for Nokia Networks' customer Vodafone.
- Installation and commission of new 3G and 4G network.

Teaching Experience

Teaching Assistant – University of Pittsburgh

- CS 1650: *Software Engineering* (Summer 20)
- CS 1566: *Computer Graphics* (Fall 20)
- CS 1550: *Intro to Operating System* (Spring 21, Summer 21)
- CS 449: *Intro to System Software* (Summer 22, Summer 23, Fall 23, Summer 24, Fall 24)
- CS 1632: *Software Quality Assurance* (Fall 22, Spring 23, Spring 24, Spring 25)
- CS 1653: *Applied Cryptography Network Security* (Summer 23)

Teaching Assistant – Stonybrook University

- CS 214: *Data Structures* (Fall 16)
- CS 114: *Introduction to Object-Oriented Programming* (Fall 17, Spring 18)
- MEC 102: *Engineering Computing and Problem Solving* – Dept of Mechanical Engineering (Spring 17, Fall 17)
- CS 102: *Introduction to Web Design and Programming* (Spring 18)

Additional Course Conducted

- Pitt Air Force Association CyberCamp – Teaching Cybersecurity ethics, principles to high school students (Summer 24, Summer 25)